

# GYAN SETU

## *Chapter: 3*

### *Pair of Linear Equations in Two Variables*

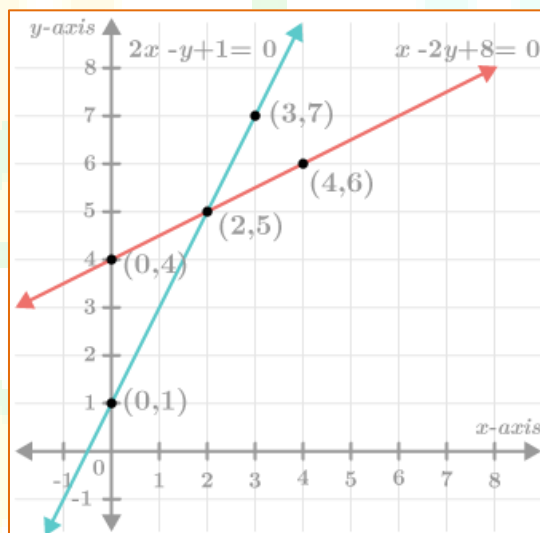
#### **Introduction**

A pair of linear equations in two variables can be solved to find unknown values.

**Example:** If  $x$  = rides,  $y$  = games,  
then equations can be formed like

$$2x - y + 1 = 0$$

$$x - 2y + 8 = 0$$



#### **Conditions for Solutions**

For equations  $a_1x + b_1y + c_1 = 0$  and  $a_2x + b_2y + c_2 = 0$ :

1.  $a_1/a_2 \neq b_1/b_2 \rightarrow$  one solution (consistent)
2.  $a_1/a_2 = b_1/b_2 \neq c_1/c_2 \rightarrow$  no solution (inconsistent)
3.  $a_1/a_2 = b_1/b_2 = c_1/c_2 \rightarrow$  infinitely many solutions (dependent & consistent)

By plotting both equations as lines:

1. **Intersecting lines** → unique solution (consistent)
2. **Parallel lines** → no solution (inconsistent)
3. **Coincident lines** → infinitely many solutions (dependent & consistent)

| S. No. | Pair of lines                           | $\frac{a_1}{a_2}$ | $\frac{b_1}{b_2}$ | $\frac{c_1}{c_2}$ | Compare the ratios                                       | Graphical representation | Algebraic interpretation      |
|--------|-----------------------------------------|-------------------|-------------------|-------------------|----------------------------------------------------------|--------------------------|-------------------------------|
| 1      | $x - 2y = 0$<br>$3x + 4y - 20 = 0$      | $\frac{1}{3}$     | $\frac{-2}{4}$    | $\frac{0}{-20}$   | $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$                   | Intersecting Lines       | Exactly one solution (unique) |
| 2      | $2x + 3y - 9 = 0$<br>$4x + 6y - 18 = 0$ | $\frac{2}{4}$     | $\frac{3}{6}$     | $\frac{-9}{-18}$  | $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$    | Coincident Lines         | Infinitely many solutions     |
| 3      | $x + 2y - 4 = 0$<br>$2x + 4y - 12 = 0$  | $\frac{1}{2}$     | $\frac{2}{4}$     | $\frac{-4}{-12}$  | $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$ | Parallel Lines           | No solution                   |

### Graphical Method

By plotting both equations as lines:

1. **Intersecting lines** → unique solution (consistent)
2. **Parallel lines** → no solution (inconsistent)
3. **Coincident lines** → infinitely many solutions (dependent & consistent)

### Algebraic Methods

1. **Substitution Method** – Express one variable in terms of the other, substitute, and solve.
2. **Elimination Method** – Multiply equations to eliminate a variable, then solve.

These methods are useful when graphing is not convenient.

**Example - Comparing Coefficients**

Check the type of solution for:

$$2x + 3y = 9$$

$$4x + 6y = 18$$

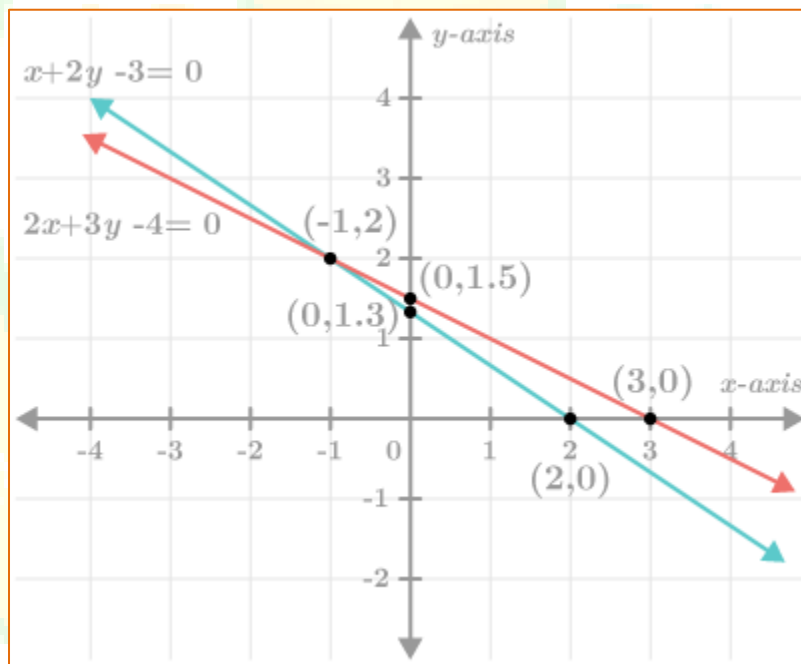
Here,  $a_1/a_2 = 2/4 = 1/2$ ,  $b_1/b_2 = 3/6 = 1/2$ ,  $c_1/c_2 = 9/18 = 1/2$ .

Since all ratios are equal, the equations are dependent  $\rightarrow$  infinitely many solutions.

**Example - Graphical Method**

Solve graphically:  $x + 2y = 3$  and  $2x + 3y = 4$

**Solution:** Plot the lines.



The two lines intersect at  $(-1, 2)$ . Hence,  $x = -1$ ,  $y = 2$ .

**Example - Substitution Method**

Solve:  $x + y = 14$  and  $x - y = 4$

**Solution:**

From  $x - y = 4$ , we get  $x = y + 4$ .

Substitute into  $x + y = 14 \rightarrow (y+4) + y = 14 \rightarrow 2y + 4 = 14 \rightarrow y = 5$ .

Then  $x = 5 + 4 = 9$ .

So, solution is (9,5).

**Example - Elimination Method**

Solve:  $2x + 3y = 11$  and  $2x - 4y = -24$

**Solution:**

Multiply 1st eqn by 2:  $4x + 6y = 22$ .

Multiply 2nd eqn by 2:  $4x - 8y = -48$ .

Subtract  $\rightarrow 14y = 70 \rightarrow y = 5$ .

Substitute into  $2x + 3y = 11 \rightarrow 2x + 15 = 11 \rightarrow 2x = -4 \rightarrow x = -2$ .

So, solution is (-2,5).

**Summary**

- Graphical & algebraic methods help solve linear equations.
- Depending on the situation, equations may have unique, no, or infinite solutions.
- These concepts are applied in real-life problems like age, money, speed-distance, etc.